

Module I

(safety introduction)

Need for safety. Safety and productivity. Definitions: Accident, Injury, Unsafe act, Unsafe Condition, Dangerous Occurrence, Reportable accidents. Theories of accident causation. Safety organization-objectives, types, functions, Role of management, supervisors, workmen, unions, government and voluntary agencies in safety. Safety policy. Safety Officer-responsibilities, authority. Safety committee-need, types, advantages.

Need for safety.

- The importance of industrial safety was realized because of the fact that every year millions occupational/ industrial accidents occur which result in loss of production time equivalent to millions of man hours, machine hours etc.
- Of these about one-fifth production time is lost by those actually injured due to temporary and permanent disablement and the remaining production time is lost by fellow operators/ people in helping the injured, in taking care of the damage caused by accident etc. the loss to the industrial unit would appear much more alarming when death cases due to accidents are considered.
- It is therefore essential to identify/examine the causes of industrial accidents and take steps to control them.
- Many disciplines are concerned with this safety approach.
- Industrial engineering is one field which deals with design of efficient work place, equipment and industrial layout design.
- Other disciplines which can contribute to safe working environment are psychology, sociology and Medicare science.
- **The following steps may be taken to effectively and efficiently eliminate an unsafe working environment:**
 1. Elimination if possible of the causes of accidents.
 2. If it is not possible to eliminate the cause of accidents, make arrangements to shield the hazardous place by guards, enclosures or similar arrangements.
- Need and concern for safety is very important.
- There are some direct costs/ effects of an accident but there are certain indirect costs involved in it also e.g. machine down time, damage to machine, ideal time of nearby equipment and horror created among workers, loss of time etc. in aid cost compensation, legal implications and allied costs etc. So safety measures would not only eliminate/ avoid above cost but would mean performing their moral responsibility towards workmen/operators also.
- To protect health of all employees
- To ensure well being of all workers
- To promote positive culture in all organization or company
- To ensure accident and incident prevention
- To ensure organization development and its growth
- To ensure continual improvement of company

Safety and productivity

- The safety of people, machines, and processes is a key element of any sustainable business or organization.

- Numerous studies show that best-in-class performers achieve higher Overall Equipment Effectiveness, less unscheduled downtime, and less than half the injury rate of average performers.
- Top manufacturers continue to discredit the age-old notion that safety and productivity have separate and competitive goals.
- These leaders use a combination of integrated safety solutions and new international standards to optimize their uptime and productivity.
- Learn how you can minimize safety-related downtime that hinders your ability to be more productive.
- Workplace productivity is dependent on employee safety. As such, employers should put safety first before focusing on productivity.
- Increasing safety in the workplace can significantly boost productivity.
- An unsafe work environment lowers productivity
- Workplace productivity refers to the amount of work accomplished by employees over a given period.
- Having a workforce that is engaged and committed can help maximize productivity.
- Productivity levels can be measured by comparing input and work output.
- The quantity and quality of work delivered by each employee can either increase or reduce productivity in the workplace.
- A productive workplace offers a wide range of benefits to the company, employer, and employees:
 - Increased Performance: Workplace performance and productivity go hand in hand. When performance is increased, so does productivity. Also, increasing productivity in the workplace can motivate employees to put in more effort, thus increasing performance.
 - Improved Employee Engagement: Efficient and effective productivity in the workplace encourages employees to be more involved in their work. Additionally, such employees are always ready to take on new tasks and often aim to be the best at what they do.
 - Greater Fulfilment: Increased productivity can give employees and employers a sense of fulfilment and purpose. Those in the workplace can be happy or unhappy based on the level of productivity.
 - Promotes a Healthy Work Culture: A productive workplace can improve employee well-being. Plus, when people achieve goals as a team, they are likely to form professional work relationships.
 - Better Revenue Generation: Long-term productivity in the workplace can maximize company profits. Organizations that are productive offer better services and can easily turn a profit.
- One of the best ways that employers can increase productivity is by creating a safe_work environment.
- Research shows that employees who feel their workplace is safe are more likely to perform better than those who feel unsafe.
- Safety is improved by removing potential hazards from the workplace.
- Aside from that, employers should create a psychologically safe environment for employees.
- A well-managed and efficient workflow, on the other hand, can improve productivity levels by eliminating redundancies when tackling tasks.
- a safe workplace with minimal job hazards is often a more productive and profitable one

Safe environments minimize lost working hours

Whether your work environment is an office or an industrial facility, reducing downtime is the key to maintaining productivity. In the office, this might mean maintaining a reliable IT infrastructure. A factory, on the other hand, would rely on having machines that run at peak efficiency. In either setting, employees are necessary to maintain optimal productivity. Because businesses rely on employees in this way, injuries will lead to lost productivity in any workplace. Businesses that cut corners to save time and speed up production can often seem productive and profitable. In the long term, these environments can also be unsafe, leading to increased downtime. While safety programs and implementing safe work practices often require an initial investment, they also have a positive impact on employee health. Because healthy employees are more reliable and productive, they're also good for the bottom line.

Safety culture at work makes for engaged employees

A key aspect of finding a balance between productivity and safety is the enforcement of safety culture. A study by Lockheed Martin of their Paducah Plant found that by developing a safety culture, they were able to increase employee productivity by 24% and reduce factory costs by 20%. Lockheed Martin's study found that the major reason for this increase was their focus on reducing errors that lead to job hazards and accidents. By implementing safe job procedures, extensive training, improving pre-work preparation, and auditing all safety processes, plant personnel were able to increase safety while also reducing expenses.

Safety information and education

As part of a safe workplace and culture, updated safety information and education, with both employees and management, is integral to success and improved safety. There are many great options for online safety training available such as the [OSHA](#) and [National Safety Council](#) in the United States and the [CCOHS](#) in Canada. But in addition to these great resources, there are many, many other options for online safety training, seminars, and education for all kinds of industries and safety challenges.

Safe work practices and safe job procedures

The Infrastructure Health and Safety Association in Ontario says:

“Companies should establish safe work practices/safe job procedures for addressing significant hazards or for dealing with circumstances that may present other significant risks/liabilities for the company. They should reflect your company's approach to controlling hazards.”

Well-researched safe work practices and safe job procedures help protect employees by providing specific instruction and directions around work performance and conduct that will reduce the safety risk of harm to employees, the environment, and equipment. These work practices and work procedures are the guidebooks to mitigating safety hazards specific to your organization and team.

Safe employees are happier and engaged in their work

Safe working conditions often go hand in hand with ideal operational conditions. With safeguards in place, employees can focus on their work, instead of the dangers and job hazards in their workplace. Keeping a cleaner, more organized space will help reduce safety hazards, and can also improve efficiency. Safety in the workplace is productive in other ways, too. Employees who take personal accountability in their safety are more likely to be engaged and happy at work. Many studies have